MILITARY SPECIFICATION

TEST SET, RADAR AN/UPM-116A

This amendment forms a part of Specification MIL-T-22619(Wep) dated 3 April 1963 and shall constitute the covering page of the specification, it has been approved by the Bureau of Naval Weapons, Department of the Navy

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3.2.7 Under Voltage Protection - The equipment shall not be damaged by operation at 21 volts and shall automatically resume normal operation when the voltage returns within limits.

3.2.8 Nomenclature and Nameplates - Nomenclature assigned and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-T-21200.

3.2.10 Maintenance Provision and Field Testing - Provision for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.11 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.12 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.3 Detail Requirements -

3.3.1 Test Set, Radar TS-1414A/UPM-116 - The radar test set shall meet the following requirements:

3.3.1.1 Function - The test set shall perform a preflight check on Radar Set AN/APQ-83 (as specified in Specification MIL-R-21642) or Radar Set AN/APQ-94 (as specified in Specification MIL-R-23373) while installed in the aircraft or on the associated test bench.

3.3.1.2 Finish - The case shall be finished with gray enamel conforming to MIL-T-21200. The panel shall be finished with black synthetic lusterless enamel in accordance with TT-E-527.

3.3.1.3 Contents of Unit - The test set shall contain the necessary circuits to perform the following functions:

3.3.1.3.1 Null Meter Voltage Measurements - The test set shall measure the following power supply voltages on a "go-no-go" basis to an accuracy of ±1%.

(1) +500 VDC
(2) +300 VDC
(3) +150 VDC
(4) -250 VDC
(5) + 20 VDC
(6) - 20 VDC

In addition, the test shall measure (7) range and (8) range rate voltages corresponding to 1000 or 17,000 yards and 360 knots (closing) on a "go-no-go" basis to an accuracy of ±1%.

3.3.1.3.2 Current Measurement - The test set shall measure the following currents to an accuracy as noted:

(1)	Magnetron Current	±5% (go-no-go)
(2)	#1 IF Crystal Current	±5% (go-no-go)
(3)	#2 IF Crystal Current	±5% (go-no-go)
(4)	#1 AFC Crystal Current	±5% (go-no-go)
(5)	#2 AFC Crystal Current	±5% (go-no-go)

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(a)	"+500"	Scaling resistors internal to radar set
(b)	"+300"	
(c)	"+150"	
(d)	"-250"	
(e)	"+ 20"	
(f)	"- 20"	
(g)	"RANGE RATE"	-2.0 VDC input signal corresponding to 360 knots closing.
(h)	"RANGE"	+85 VDC input signal corresponding to 17,000 yards or +5 VDC input signal corresponding to 1000 yards as selected by the RANGE CAL switch.

(5) Crystal Current Meters - Four one-inch sealed and ruggedized panel meters shall be provided for continuously monitoring the four crystal currents. They shall have 0-100 microamp movements. The scales shall be marked with parameters indicating 0.7 ma (min) and 1.2 ma (max) and shall be calibrated with an accuracy of ±2% of full scale deflection at the 0.9 ma scale division.

(6) Antenna Position Switch - The antenna position switch shall switch the modulation input voltages and provide a means of energizing the scan failure relay in the Radar Set and shall have the following labeled positions:

(a) "SCAN FAIL"

(b) "OFF"

(c) "UP"

(d) "RIGHT"

(e) "DOWN"

(f) "LEFT"

(7) Range Targets Switch - A switch labeled "RANGE TARGETS" "ON-OFF" shall be provided to connect +28 VDC to the test targets relay in the Radar Set.

(8) Meter Sensitivity Switch - A switch labeled "METER SENSITIVITY" "MAX-MIN" shall be provided to protect the Null Meter against overloads. This is a spring loaded toggle switch which is normally in the Min. sensitivity position. Operation of this switch to the "MAX" position removes a shunt resistor from the Null Meter circuit.

(9) Antenna Test Switch - A switch labeled "TEST" "AGC-OFF-TRACK" shall be provided to connect +28 VDC to the Angle Track Test Relay and the AGC Test Relay in the Radar Set.

(10) Transmitter Disable Switch - A switch labeled "XMTR" "OFF-NORMAL" shall be provided to connect +28 VDC to the transmitter disable relay in the Radar Set and at the same time to turn on the 30 mc pulsed oscillator in the test set.

(11) Range Rate Test Switch - A switch labeled "RANGE RATE TEST" shall be provided to operate in conjunction with the multimeter and multimeter function switch in the neg. range rate (closing) position and shall have the following labeled positions:

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(a) "TARGET RATE" - Sets multimeter scale factor for -2.78 VDC full scale representing 500 knots closing.

(b) "STROBE RATE" - Sets multimeter scale factor for +40 VDC full scale for range strobe rate check.

(c) "VELOCITY TEST" - Functions same as (a) but also connects voltage to Radar Set range rate test circuit.

(12) Range Rate Control - A control labeled "RANGE RATE" "INCREASE-DECREASE" shall be provided to adjust the level of the DC voltage connected to the range rate section of the range computer. The range of the control shall be such that target rates between 300 and 500 knots closing can be simulated. The voltage output of this control shall be between a minimum of +0.73 ±0.1 VDC and a maximum of +1.40 ±0.07 VDC when connected to a 500 ohm ±0.1% resistive load.

(13) Tune Control - A control labeled "TUNE" "AUTO-MANUAL" shall be provided to allow manual adjustment of the klystron repeller voltage and shall have a range of from -78 ±5.0 VDC to -158 ±10 VDC. This control shall be provided with a switch which shall connect +28 VDC to the Radar Set auto-manual tune relay except when the control is in its most CCW position.

(14) Modulation Control - A control labeled "MODULATION" shall be provided to adjust the level of the 24 cycle modulation signal. This control shall operate from a source impedance of 600 ohms ±10%.

(15) Range Cal Switch - A switch labeled "RANGE CAL" "1000 YARDS - 17000 YARDS" shall be provided to select the range function of the null meter.

3.3.1.4.2 Connectors -

(1) Front Panel Connectors - The following front panel connectors shall be provided.

(a) Spin Frequency - A UG-657(*)/U BNC connector labeled "SPIN FREQ" shall be provided to allow measurement or use of the spin frequency reference. This signal is obtained from the output of the modulation control.

(b) System Trigger - A UG-492(*)/U BNC connector labeled "SYSTEM TRIGGER" shall be provided to connect the Radar Set system trigger to the 30 mc pulsed oscillator.

(c) Test Targets Trigger - A UG-492(*)/U BNC connector labeled "TEST TARGET TRIGGER" shall be provided to connect the 30 mc pulsed oscillator output to the Radar Set.

(2) Case Connectors - Two multi-pin connectors shall be provided on the side of the case to provide test connections to the aircraft test panel or the test bench harness. The connectors shall be identified as J101 and J102.

3.3.2 Cable Assembly, Special Purpose W-1 of AN/UPM-116 - The equipment shall be furnished with a 10-foot test cable to provide connections to the aircraft test panel. The connectors shall be identified as P101, P102, P2001, P2004, P2005, P2006, P2007 and P2009.

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